Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Nadeau, Jay L Ph.D.		
Project Title:	Miniature Electronic Dynamic Ion Channel Sensor (MEDICS)		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA JPL		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8099	**Congressional District:**	29
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	07/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are designing and fabricating a stable, fully-enclosed biosensor that uses ion channel proteins as the signal transduction mechanisms. The device consists of 4 or 8 individually-adressable wells in which ion channels are assembled and which yield DC current measurements to an attached amplifier. The goals are two-fold: first, to perform basic molecular biology and mutagenesis to create ion channels that are useful in sensors for life detection and health monitoring. Second, to develop these engineered channels into stable biosensors, using MEMS technology and biomimetic lipid membranes. The advantages of the current design over existing ion channel biosensors are (1) increased stability due to the agarose layer; (2) reduced electrical noise due to the thick dielectric, which is constructed from a particular photoresist; (3) ability to incorporate any type of channel, not only gramicidin; and (4) the use of a large reservoir, which allows for DC current measurements. Ion channels yield very sensitive and specific signals to a wide array of ligands. They thus would be ideal bases for biosensors if they could be stably expressed and addressed.		

	However, ion channels only function within biological or "biomimetic" membranes, whose lifespan is usually several hours or less. Additionally, their small signals (1-10 picoamps) make a low-noise, highly amplified detection platform necessary. Because of these complications, ion channels have not yet been made into commercial devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)